Python Team Progress Report: Environment & Integration

Project: Ride-Share Demand Analysis (`ride-share-demand-analysis`)

Python Members:

Environment: Google Colab + GitHub

Date: April 1, 2026

1. Development Environment Setup

- **Notebook Initialization:** Created and standardized the primary analysis notebook: `EDA_Ride_Share_Demand_Analysis.ipynb`.
- **Version Control:** Established a direct "Save to GitHub" workflow. The notebook is now tracked in the team repository under the `/notebooks` directory.
- **Authentication:** Successfully implemented the `google.colab.auth` module to allow secure, programmatic access to our Google Cloud Project.

2. BigQuery Python Client Integration

We have successfully initialized the `google.cloud.bigquery` library. The Python environment is now capable of executing complex SQL strings and returning the results directly as **Pandas DataFrames** for analysis.

Core Connection Script:

```python
from google.cloud import bigquery
from google.colab import auth

# 1. Authenticate User
auth.authenticate_user()

# 2. Initialize the BigQuery Client
project_id = 'ride-share-demand-analysis'
client = bigquery.Client(project=project_id)
```

3. Data Loading & Optimization

We have transitioned from basic data previews to **Optimized Data Retrieval**. To stay within our 10 GB student quota, we are moving away from `SELECT *` and instead targeting specific features.

- **Initial Test Case:** January 2023 Preview.
- **Optimization Result:** By selecting only required columns (e.g., `fare`, `trip_start_timestamp`), we reduced our DataFrame memory footprint and BigQuery scan weight by over **25%** (from 73 GB down to ~53 GB for a one-month slice).
- **Method:** Using the `.to_dataframe()` method to convert SQL results into a structured format ready for `matplotlib` and `seaborn` visualization.

4. Python Team Checklist (Week 1 & 2 Status)

Task	Status	Notes
Colab Connectivity	☑ **Complete**	Client is successfully pulling 2023 data.
Library Imports	☑ **Complete**	Pandas and BigQuery clients are active.
DataFrame Validation	☑ **Complete**	Confirmed `df.head()` renders correctly.
Data Cleaning Logic	🕒 **In Progress**	Writing logic to handle `$0` fare outliers.
GitHub Sync	☑ **Complete**	Notebook is saved in the team repo.

Next Steps for Python Team:

- **Week 2 EDA:** Generate time-series plots for hourly demand trends.
- **Statistical Analysis:** Calculate the standard deviation and variance of fares for the 2023 fiscal year.
- **Visualization:** Build the initial "Top 10 Pickup Areas" bar chart using Seaborn.